ECE 447

Lesson 34 – GR2

Review



Schedule and Admin

- Schedule.
  - Lesson 34 - **GR2 Review Day** (today)
  - Lesson 35 - **GR2** (Tuesday, 17 Nov) – covering Lessons 18-32, Chapters 5-8
  - Lessons 36-38 - Binary digital system performance, error correction
  - Lesson 39 - SDR Lab 8: capstone image transfer
  - Lesson 40 - Advanced topics: OFDM, MIMO, CDMA
  - Lesson 41 - Course review
- Admin
  - **HW8**. Due tonight (Lesson 34, 13 Nov) to Gradescope – Chapters 7-8, 55 points. Bring your questions to today's session.
  - **HW7**. Was due Lesson 32 (Fri, 6 Nov).
  - **Lab 7**. Due Lesson 37 (Mon, 23 Nov).
  - EI is available – Outlook calendar invite, any open time.

GR2 Logistics

- **When:** Lesson 35, one class period (53 minutes). Budget your time – roughly a minute per point.
- **Coverage:** Lessons 18-32, **Chapters 5-8** — sampling, PCM, TDM, line coding, ISI and pulse shaping, M-ary, digital carrier modulation, probability, Gaussian detection, and **random processes** (Lesson 32). **Not** tested: Chapter 9 – that starts at Lesson 36.
- **Bring:** calculator and a pencil. You get the **course reference sheet** and a Q**-function table**.
- R_b **vs.** R_s**.** The GR will **clearly** distinguish bit rate from symbol (baud) rate. Read which one the problem hands you.
- **The Nyquist rate is** $2B$, not B — the most expensive place on this GR to slip.
- **Show your work**, box your answers, carry units, and use engineering notation.

A **working session**: five problems, 100 points, a full 53-minute GR. Each reinforces a homework set — **Problems 4–5 track HW8, due tonight**. Work them first, then ask.

Recap: Sampling, PCM, and TDM (Ch. 5, Lsn 18-20)

- **Sampling.** Multiplying by an impulse train in time convolves with impulse train in frequency: $\overline{M}(f) = f_s \sum_n M(f - nf_s)$ – scaled copies of the message spectrum every f_s. The **Nyquist rate is** $2B$. Sample below it and the copies overlap: **aliasing**, which is irreversible. A tone at f_0 folds to $|f_0 - f_s|$. The only fix is an anti-alias filter *before* the sampler.
- **Reconstruction.** sinc $(\cdot)$ is the ideal interpolator but is not realizable. Any real pulse $p(t)$ filters the samples, so you follow it with an **equalizer** that inverts $P(f)$ over the message band, is flexible in the guard band, and is zero beyond $f_s - B$. Sampling above Nyquist is what buys that guard band.
- **PCM.** Sample, quantize, encode. With $L = 2^n$ levels over $\pm m_p$, the step is $\Delta = 2m_p/L$ and the *maximum* error is $\Delta/2 = m_p/L$. Each added bit buys about **6 dB** of SNR.
- **Bandwidth.** Nyquist's noiseless limit is 2 bps/Hz, so $B_T = R_b/2$ with $R_b = nf_s$. *Only* when sampling exactly at Nyquist does this collapse to $B_T = nB$ – check before using the shortcut.
- **TDM.** N channels interleave into Nnf_s bits/s, so bandwidth scales with N. Going the other way, a fixed allocation carries $\lfloor 2B_T/R_b \rfloor$ channels – round **down**.

Recap: Line Codes and ISI (Ch. 6, Lsn 22-24)

- **Line codes.** $y(t) = \sum_k a_k p(t - kT_b)$: the code picks the amplitudes a_k, the pulse shape sets the spectrum. Judge one on bandwidth, power efficiency, DC content, self-clocking, error detection, transparency. Polar NRZ is cheap but has no DC null and no guaranteed transition; Manchester gets both, at roughly double the bandwidth.
- **ISI.** Not noise – it is what a channel that is not distortionless over the signal band does to pulses. Time-limited implies infinite bandwidth; band-limited implies infinite duration. Either way pulses spread past their T_b slot, which is tolerable as long as the spreading is zero *at the decision instant*.
- **Nyquist's First Criterion.** $p(0) = 1$ and $p(\pm nT_b) = 0$. Pulses may overlap freely *between* decision instants – all that matters is that every other pulse contributes exactly zero at each sampling point. $\text{sinc}\,(\pi R_b t)$ meets it with the minimum bandwidth $R_b/2$, but its $1/t$ tails make it hopelessly sensitive to timing error – hence **raised cosine**.

Recap: Pulse Shaping and M-ary (Ch. 6, Lsn 24-27)

- **Raised cosine.** $B_T = \frac{(1+r)R_b}{2}$ with roll-off $r \in [0, 1]$; the excess bandwidth is $f_x = \frac{rR_b}{2}$, the part above the $R_b/2$ Nyquist minimum. More r costs bandwidth and buys timing robustness. Watch the factor of two – B_T is *half* of $(1 + r)R_b$.
- **M-ary.** $n = \log_2 M$ bits per symbol, so $R_s = R_b/n$ and $B_T = \frac{(1+r)R_s}{2}$. Going M-ary is how you fit a fixed bit rate into a narrower band. Not free – packing M levels into the same amplitude range shrinks the noise margin and costs transmit power (about $5\times$ for 4-ary PAM, per Example 6.6).
- **Eye diagrams.** Eye *width* is your timing-jitter margin; eye *height* is your noise margin. An M-ary eye has $M - 1$ openings stacked vertically, each correspondingly smaller.
- **Carrier modulation.** ASK, PSK ($\varphi_{PSK} = m(t) \cos \omega_c t$ with $m(t)$ polar, so the two phases sit π apart), FSK, and QAM. The modulation property gives the bandpass PSD: $S_\varphi(f) = \frac{1}{4} S_M(f + f_c) + \frac{1}{4} S_M(f - f_c)$.

Recap: Probability (Ch. 7, Lsn 30)

- **Keep the three straight.** Joint $P(X, Y)$, conditional $P(Y \mid X)$, marginal $P(Y)$. Mixing them up is the most common way to lose a whole problem. **Total probability** walks conditionals up to a marginal, $P(Y) = \sum_X P(Y \mid X)P(X)$; **Bayes** runs it backwards, $P(X \mid Y) = \frac{P(Y|X)P(X)}{P(Y)}$.
- **Binary symmetric channel.** $P(Y \neq X) = p$ either direction, so $P(Y = 1) = P(X{=}1)(1 - p) + P(X{=}0)p$. Free check: your marginals must sum to 1. And the source priors do *not* have to be 0.5 – read the problem.
- **Bernoulli trials.** n independent bits at error probability p: mean np, variance $np(1 - p)$, and $P(\text{no errors}) = (1 - p)^n$. Do not approximate that last one by $1 - np$ unless you know $np \ll 1$ – it is the union bound, and it drifts badly as p grows.

Recap: Gaussian RVs and RPs (Ch. 7-8, Lsn 31-32)

- **Gaussian RV.** $p_x(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-(x-m)^2/2\sigma^2}$, with tail area $P(\mathrm{x} > x) = Q\left(\frac{x-m}{\sigma}\right)$. The argument of Q is a *number of standard deviations* – dimensionless. Never write Q of a voltage.
- **Detection.** $r(T_b) = s_k(T_b) + n(T_b)$ with n Gaussian. For equally likely $\pm A$ the optimum threshold is $a_0 = 0$ and $P_e = Q\left(\frac{A}{\sigma}\right)$. Unequal priors shift the threshold toward the *less* likely symbol. Since Q falls like $e^{-x^2/2}$, a small change in A/σ moves P_e by orders of magnitude – and P_e depends on that *ratio* alone, which is the whole link-budget argument.
- **Moments.** Mean $\mathbb{E}[\mathrm{x}]$, mean square $\overline{\mathrm{x}^2}$, variance $\sigma^2 = \overline{\mathrm{x}^2} - \overline{\mathrm{x}}^2$. Means always add; variances add only for *independent* RVs. **CLT:** sums of many independent RVs go Gaussian regardless of their own PDFs.
- **Random processes.** An ensemble of sample functions rather than of numbers. WSS means the mean is constant and $R_x(t_1, t_2)$ depends only on τ; then $R_x(\tau) \leftrightarrow S_x(f)$ and $P_x = R_x(0) = \int S_x(f)\, df$.

Problem 1: Sampling and Aliasing [20 pts]

An audio signal $m(t)$ is band-limited to $B = 4\,\text{kHz}$ and is sampled at **25% above the Nyquist rate**.

- **(a)** [4] Find the Nyquist rate, the sampling rate f_s, and the sampling interval T_s.
- **(b)** [5] Sketch the spectrum of the sampled signal. Label the replica spacing, the amplitude scaling, and the width of the guard band.
- **(c)** [5] A 9 kHz tone leaks past the input filter and is sampled along with $m(t)$. Where does it land? Can you remove it afterwards? What is the correct fix, and where in the chain does it go?
- **(d)** [6] Reconstruction uses the flat-top pulse $p(t) = \Pi\left(\frac{t}{\tau}\right)$ with $\tau = T_s/2$. Find the required equalizer frequency response $E(f)$, including all three bands.

Problem 2: PCM, Quantization, and TDM [20 pts]

A voice channel band-limited to $B = 4\,\text{kHz}$ is sampled at $f_s = 10\,\text{kHz}$ and PCM encoded.

- **(a)** [4] The maximum quantization error must be no more than 0.2% of the peak amplitude m_p. Find the minimum number of bits per sample n.
- **(b)** [4] You upgrade from 9 to 12 bits per sample. How much does the quantization SNR improve, in dB and as a power ratio? What does it cost?
- **(c)** [5] With $n = 9$ bits per sample and $f_s = 10\,\text{kHz}$, find the bit rate R_b and the minimum channel bandwidth B_T for a noiseless channel.
- **(d)** [7] A $1.0\,\text{MHz}$ channel is available. Using the $2\,\text{bps/Hz}$ Nyquist limit, how many of these $90\,\text{kbps}$ PCM voice channels can be time-division multiplexed onto it? Justify how you round.

Each part restates what it needs – a slip in (a) will not cost you (c) or (d).

Problem 3: Line Coding and ISI [22 pts]

- **(a)** [6] A source sends **1 0 1 1 0** at $R_b = 1\,\text{Mbps}$. Sketch $y(t) = \sum_k a_k p(t - kT_b)$ for (i) polar NRZ and (ii) Manchester, labeling T_b. Which has DC content? Which is self-clocking, and what does that cost?
- **(b)** [4] State Nyquist's First Criterion for zero ISI. $\text{sinc}(\pi R_b t)$ satisfies it at the minimum possible bandwidth – so why is it not used in practice?
- **(c)** [6] Binary polar signaling at $R_b = 1\,\text{Mbps}$ uses a raised-cosine pulse with roll-off $r = 0.5$. Find B_T and the excess bandwidth f_x.
- **(d)** [6] Your allocation is cut to $B_T = 450\,\text{kHz}$, but R_b must stay at 1 Mbps. **(i)** Show that no binary scheme can do this. **(ii)** Switch to 4-ary PAM: find R_s and the required r. What is the cost?

Problem 4: Probability and the BSC [18 pts]

A binary symmetric channel has crossover probability $p = 0.01$. The source is **not** equiprobable: it sends $X = 0$ with probability 0.7 and $X = 1$ with probability 0.3.

- **(a)** [5] Find $P(Y = 0)$, the probability that the received bit is a 0.
- **(b)** [5] You receive a 1. How certain are you that a 1 was sent? Find $P(X = 1 \mid Y = 1)$.
- **(c)** [4] A 1000-bit packet crosses the same channel. Find the expected number of bit errors and the variance.
- **(d)** [4] Find the probability that a 10-bit word arrives with no errors at all. Comment on what that implies about word error rate.

Problem 5: Gaussian Detection [20 pts]

A polar baseband system samples its receive filter once per bit:

$$r(T_b) = \pm A + n(T_b), \qquad n(T_b) \sim \mathcal{N}(0, \sigma^2)$$

with $+A$ sent for a 1 and $-A$ for a 0, each equally likely.

- **(a)** [5] Sketch the two conditional PDFs and mark the optimum decision threshold a_0. Justify where it sits, and shade the area that represents P_e.
- **(b)** [5] With $A = 3\,\text{V}$ and $\sigma = 1\,\text{V}$, find P_e using the Q function and the table.
- **(c)** [5] The noise standard deviation rises to $\sigma = 1.5\,\text{V}$ with A unchanged. Find the new P_e. By what factor did it change?
- **(d)** [5] To restore the original P_e at $\sigma = 1.5\,\text{V}$, what amplitude A is needed, and by what factor must the transmit power increase?